Cisco PE Node MALL-PE-08 Standard Operating Procedure (SOP) for Optical Transport Failure (ALARM_7699) - Warning Severity

Node: MALL-PE-08 | **Vendor:** Cisco | **Severity:** Warning | **Alarm Code:** ALARM_7699

Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution steps for Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on Cisco PE Node MALL-PE-08 at the Malleswaram Node, Role: PE. The failure domain is typically triggered by degradation of SFP/XFP optical transceivers, fiber cuts, or loss of signal strength in the receive (Rx) or transmit (Tx) direction. The underlying protocol or hardware mechanics involve the Cisco PE router's ability to detect and report optical transport failures through the Optical Interface Card (OIC), which is responsible for converting electrical signals to optical signals and vice versa. The OIC is equipped with a dedicated microprocessor that continuously monitors the optical signal strength, temperature, and other parameters. When an optical transport failure is detected, the OIC generates an alarm (ALARM_7699) and sends a trap notification to the Cisco PE router, which in turn reports the failure to the Network Management System (NMS).

Pre-requisite Checks

- Ensure dispatch of remote hands to the Malleswaram Node to physically inspect the affected equipment.

- Verify the status of redundant links to ensure they are operational and can take over traffic in case of a failure.

- Gather information on the affected SFP/XFP optical transceiver, including its serial number, type, and revision.

- Review the event log and alarm history to identify any prior occurrences of similar failures.

Potential Root Causes

Cause	Probability
Degraded SFP/XFP optical transceiver	High
Fiber cut or damage	Medium
Loss of signal strength in Rx or Tx direction	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface optics MALL-PE-08/0/0, Expected Output: Optics information for interface MALL-PE-08/0/0, Description: This command displays the current optics information, including the SFP/XFP type,
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serial number, and optical signal strength.

Command: show run | include alarm, Expected Output: Alarm
configuration and status, Description: This command displays the
current alarm configuration and status, including the ALARM_7699
notification.

Command: debug optics, Expected Output: Optics debug information,
Description: This command enables optics debugging, which provides
detailed information on the optical signal strength, temperature, and
other parameters.
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Resolution Steps

Step 1: Step: Replace the degraded SFP/XFP optical transceiver, Command: no interface optics MALL-PE-08/0/0, Description: This command removes the current optics configuration and allows for the replacement of the degraded SFP/XFP optical transceiver.

Step 2: Step: Verify the new SFP/XFP optical transceiver, Command: show interface optics MALL-PE-08/0/0, Description: This command verifies the current optics information and ensures the new SFP/XFP optical transceiver is properly configured and operational.

Step 3: Step: Clear the alarm and notification, Command: clear alarm, Description: This command clears the ALARM_7699 notification and removes any associated alarm configuration.

Post-Check Verification

```
Command: show interface optics MALL-PE-08/0/0, Expected Output: Optics
information for interface MALL-PE-08/0/0, Description: This command
verifies the current optics information and ensures the new SFP/XFP
optical transceiver is properly configured and operational.

Command: show run | include alarm, Expected Output: Alarm
configuration and status, Description: This command verifies the
current alarm configuration and status, including the ALARM_7699
notification.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP optical transceiver, Command: interface optics MALL-PE-08/0/0, Description: This command restores the original optics configuration and ensures the degraded SFP/XFP optical transceiver is re-installed.

- Step: Clear the alarm and notification, Command: clear alarm, Description: This command clears the ALARM_7699 notification and removes any associated alarm configuration.

- Step: Verify the original SFP/XFP optical transceiver, Command: show interface optics MALL-PE-08/0/0, Description: This command verifies the current optics information and ensures the original SFP/XFP optical transceiver is properly configured and operational.